

## 6. FACILITIES AND TECHNICAL SUPPORT (80)

### 6.1 Adequate and well-equipped laboratories, and technical manpower: (30)

| Sl. No. | Name of the Laboratory                                                                                                                                                                                                                                                 | No. of students per setup(Batch Size) | Name of the important equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Weekly utilization status (all the courses for which the lab is utilized) | Technical Manpower Support  |                |                                             |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------|----------------|---------------------------------------------|
|         |                                                                                                                                                                                                                                                                        |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                           | Name of the technical staff | Designation    | Qualification                               |
| 1       | <p>CONCRETE AND HIGHWAY ENGINEERING LABORATORY (146 Sq.m)</p> <p>(i) CE3412 Materials Testing Laboratory (odd) (4 Hours / Week)</p> <p>(ii) CE3511 Highway Engineering Laboratory (Odd) (4 Hours / Week)</p> <p>(iii) CE8811 Project Work (Even) (34 Hours / Week)</p> | 3                                     | Concrete Mixer, Slump Cone, Flow Table ( Motorized ), Vibrating Table, Compression Testing Machine 2000KN, Vee Bee Consistometer, Aggregate Impact Testing Machine, CBR Apparatus, Blains Apparatus, Los Angels Abrasion Testing Machine, Marshall Stability Apparatus, Compressometer, Compaction Factor Test Apparatus, Sieve Shaker, Universal Penetrometer, Softening (Ring & Ball Apparatus ), Ductility With Elastic Recovery Mould, Flash & Fire Point, Viscometer, Pycnometer, Needle Vibrator, Mechanical Extensometer, Film Stripping Device, Cylindrical Metal Measures, Rebound Hammer, RCPT Apparatus. | 42 Hours (Odd Semester - 8 hours, Even Semester - 34 hours)               | Mr.Nagaraj                  | Lab Technician | Bachelor of Engineering (Civil Engineering) |
| 2       | <p>ENVIRONMENTAL ENGINEERING LABORATORY/WATER AND WASTE WATER ANALYSIS LABORATORY (109 Sq.m)</p> <p>(i) CE3311 Water and Wastewater Analysis Laboratory (Odd ) (3 Hours / Week)</p> <p>(ii) CE8811 Project Work (Even) (34 Hours / Week)</p>                           | 3                                     | BOD Analyzer, COD Digester, Spectrophotometer, Flame Photometer, Gas Chromatography, Autoclave, Bacteriological Incubator, BOD Incubator, Sterilization Chamber                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 37 hours (Odd Semester - 8 hours, Even Semester - 34 hours)               | Mr.A.Arumuga Ganesh         | Lab Technician | Diploma in Civil Engineering                |

| Sl. No. | Name of the Laboratory                                                                                                                              | No. of students per setup(Batch Size) | Name of the important equipment                                                                                                                                                                                                                                                                                                                                                                                                 | Weekly utilization status (all the courses for which the lab is utilized) | Technical Manpower Support  |                |                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------|----------------|-----------------------------------|
|         |                                                                                                                                                     |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                           | Name of the technical staff | Designation    | Qualification                     |
| 3       | SOIL MECHANICS LABORATORY (105 Sq.m)<br>(i) CE3413 Soil Mechanics Laboratory (Even) (3 Hours / Week)                                                | 3                                     | Triaxial Test Apparatus, Direct Shear Apparatus, Unconfined Test Apparatus, Standard Penetration test Apparatus                                                                                                                                                                                                                                                                                                                 | 3 Hours (Even Semester - 3 Hours)                                         | Mr.V.Krishna kumar          | Lab Technician | B.Tech. in Civil Engineering      |
| 4       | FLUID MECHANICS & MACHINERY LABORATORY/HYDRAULIC ENGINEERING LABORATORY (181 Sq.m)<br>CE3411 Hydraulic Engineering Laboratory (Even) (3 Hours/Week) | 3                                     | Submersible Pump Test Rig, Orifice and Mouthpiece Setup, Rotameter Test Rig, Orifice and Venturimeter Setup, Pipe Friction Apparatus(Major Losses), Pipe Friction Apparatus(Minor Losses), Francis Turbine Test Rig(Electrical Loading), Kaplan Turbine Test Rig(Electrical Loading), Pelton Wheel Turbine Test Rig, Centrifugal Pump Test Rig, Reciprocating Pump Test Rig, Vane and Gear Pump Test Rig, Bernoullis Apparatus. | 3 Hours (Even Semester - 3 Hours)                                         | S.Manibharathi              | Lab Technician | Diploma in Civil Engineering      |
| 5       | SURVEYING LABORATORY (68 Sq.m)<br>(i) CE3361 Surveying and Levelling Laboratory (Odd) (3 Hours / Week)                                              | 4                                     | Total Station, Theodolite, Dumpy Level, Prismatic Compass, Surveyor Compass, LDM, GPS                                                                                                                                                                                                                                                                                                                                           | 3 Hours (Odd Semester - 3 Hours)                                          | Mr.K.Mareeswaran            | Lab Technician | Diploma in Civil Engineering      |
| 6       | Strength of Materials Laboratory (124 Sq.m)<br>(i) CE3412 Materials Testing Laboratory (odd) (4 Hours / Week)                                       | 3                                     | Universal Testing Machine, Torsion Testing Machine, Impact (Izod/Charpy) Testing Machine, Rockwell Hardness Testing Machine, Brinell Hardness Testing Machine, Spring Testing Machine, Deflection Testing Machine with dial gauge,                                                                                                                                                                                              | 3 Hours (Odd Semester - 3 Hours)                                          | P. Murugesan                | Lab Technician | Diploma in Mechanical Engineering |

| Sl. No. | Name of the Laboratory                                                                                                                                                                                                                                                        | No. of students per setup(Batch Size) | Name of the important equipment                                                                                                                                                                                                                                                                                                                                                 | Weekly utilization status (all the courses for which the lab is utilized) | Technical Manpower Support  |                |                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------|----------------|---------------------------------------------|
|         |                                                                                                                                                                                                                                                                               |                                       |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                           | Name of the technical staff | Designation    | Qualification                               |
|         |                                                                                                                                                                                                                                                                               |                                       | Longitudinal Compressometer, Extensometer, Le Chatelier Apparatus, Vicats apparatus,                                                                                                                                                                                                                                                                                            |                                                                           |                             |                |                                             |
| 7       | Construction Practices Lab<br>(66 Sq.m) (Odd) ( 3 Hours / Week), (Even) (2 Hours/Week)                                                                                                                                                                                        | 3                                     | Adjustable Telescopic Prop - Normal Head, Adjustable Telescopic Prop - Beam Head, Span - 8', Floor Farms - Normal (900 X 600), Floor Farms - Adjustable (900 X 150), General Purpose Scaffolding, H - Frame Cup Lock Scaffolding.                                                                                                                                               | 5 Hours (Odd Semester - 3 Hours, Even Semester – 2 Hours)                 | Mr.Nagaraj                  | Lab Technician | Bachelor of Engineering (Civil Engineering) |
| 8       | Centre for Geospatial Technology (110 Sq.m) (Even) (2 Hours/Week)                                                                                                                                                                                                             | 3                                     | Differential Global Positioning system, Avio-M1 Professional Mapping Drone with PPK system, IntelCore/i7Processor/14700vPro/64 GB;7200rpm harddisk drive- 24” monitor,                                                                                                                                                                                                          | 3 Hours (Even Semester - 3 Hours)                                         | Mr.K.Mareeswaran            | Lab Technician | Diploma in Civil Engineering                |
| 9       | CAD – DIGITALIZED CONSTRUCTION LABORATORY (110 Sq.m)<br><br>(i) CE8711 Creative and Innovative project (Odd) (4 Hours / Week)<br>(ii) Building Information Modelling (Odd) (5 Hours / Week)<br>(iii) CE3611 Building Drawing and Detailing Laboratory (Even) (4 Hours / Week) | 1                                     | (i) Dell OptiPlex 3020-Intel Core/ i3-4130 Processor/4GBMemory/500GB Hard DD/Keyboard and Optical Mouse / DVD RW /18.5" Monitor.<br>(ii) HP LaserJet 1020 Plus Printer<br>(iii) AutoCAD 2023<br>(iv) Bentley STAAD.Pro Academic Software- 5 Users Perpetual Licenses Configurable up to 100 Users of Student Server Licenses.<br>(v) Primavera Contractor P6 (Latest) – Product | 14 hours (Odd Semester - 9 hours, Even Semester - 5 hours)                | Mrs.M.Thirumalai Selvi      | Lab Technician | B.Com.                                      |

| Sl. No. | Name of the Laboratory                                      | No. of students per setup(Batch Size) | Name of the important equipment                                                                                                     | Weekly utilization status (all the courses for which the lab is utilized) | Technical Manpower Support  |             |               |
|---------|-------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------|-------------|---------------|
|         |                                                             |                                       |                                                                                                                                     |                                                                           | Name of the technical staff | Designation | Qualification |
|         | (iv) Building Information Modelling (Even) (1 Hours / Week) |                                       | (vi)Epson S31 Projector<br>(vii) ETABS Ultimate V-17-10 Users Education and Research License<br>(viii) Bently BIM Software (Latest) |                                                                           |                             |             |               |



**Fig.6.1.1 CONCRETE AND HIGHWAY ENGINEERING LABORATORY**



**Fig.6.1.2 ENVIRONMENTAL ENGINEERING LABORATORY/WATER AND WASTE WATER ANALYSIS LABORATORY**



**Fig.6.1.3 SOIL MECHANICS LABORATORY**



**Fig.6.1.4 SURVEYING LABORATORY**





**Fig6.1.5 FLUID MECHANICS & MACHINERY LABORATORY/HYDRAULIC ENGINEERING LABORATORY**



**Fig6.1.6 CAD – DIGITALIZED CONSTRUCTION LABORATORY**



**Fig6.1.7 Centre for Geospatial Technology**





**Fig6.1.8 Strength of Materials Laboratory**



**Fig6.1.9 Construction Practices Laboratory**



**Fig.6.1.10 Project & Consultancy Laboratory**

## 6.2 Additional facilities created for improving the quality of learning experience in laboratories: (25)

| Sl. No                                         | Facility Name             | Details                                                                       | Reason(s) for creating facility                                                                                           | Utilization                     | Areas in which students' are expected to have enhanced learning                                                                           | Relevance to POs/PSOs                                                            |
|------------------------------------------------|---------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1. CAD – DIGITALIZED CONSTRUCTION LABORATORY   |                           |                                                                               |                                                                                                                           |                                 |                                                                                                                                           |                                                                                  |
| 1.                                             | ETABS software            | ETABS Ultimate V-17-10 Users Education and Research License                   | ✓ Students are able to analysis and design the structure.                                                                 | For Projects & Consultancy work | ✓ To enhance the technical knowledge in analyzing and design the structure.<br>✓ To enhance the technical knowledge in Numerical Studies. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 2.                                             | Primavera software        | Primavera P6 Project planning software, 75 Users                              | ✓ Students are able to use this software for project planning, Scheduling, progress tracking and resource management.     |                                 |                                                                                                                                           |                                                                                  |
| 3.                                             | Bentley BIM Software 2024 | Bentley BIM Software refers to a suite of tools developed by Bentley systems, | ✓ Students are able to enhance project visualization and coordination, also improve the design accuracy and efficiency    |                                 |                                                                                                                                           |                                                                                  |
| 4.                                             | Ansys                     | Ansys Structural FEA (Finite Element Analysis) 60 Users                       | ✓ Students are capabilities for linear and nonlinear structural analysis, including static, dynamic and thermal analyses. |                                 |                                                                                                                                           |                                                                                  |
| 5.                                             | Abaqus 6.14               | Abaqus 6.14 – SIMULIA – Finite Element Software 60 Users                      | ✓ Students are capabilities for linear and nonlinear structural analysis, including static, dynamic and thermal analyses. |                                 |                                                                                                                                           |                                                                                  |
| 2. CONCRETE AND HIGHWAY ENGINEERING LABORATORY |                           |                                                                               |                                                                                                                           |                                 |                                                                                                                                           |                                                                                  |
| 6.                                             | Rebound Hammer            | To Perform Non-Destructive Tests on concrete and other                        | ✓ Students are able to understand the procedure of conducting non                                                         | For Project & Consultancy work  | Non Destructive Testing (NDT) Studying the compressive strength of a                                                                      | PO1, PO2, PO3, PO4, PO5, PO6,                                                    |



| Sl. No | Facility Name         | Details                                                                                                                                                                              | Reason(s) for creating facility                                                                                                                                  | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                             | Relevance to POs/PSOs                                                            |
|--------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                       | specimens, existing structural components, etc.                                                                                                                                      | destructive testing for their project work and to test the specimens without braking them and also they learn to do the tests in existing structural components. |                                | specimen or a building component without causing damage and to find out the rate of development of compressive strength of concrete cubes.                                                  | PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4                               |
| 7.     | RCPT Apparatus        | To perform durability tests on concrete and particularly to find out the chloride penetration.                                                                                       | ✓ Students are able to do projects on durability properties of concrete with various partial replacements and also useful for carrying out researches.           | For Project & Consultancy work | Durability Studies<br>Acquire knowledge in durability study of concrete and to conduct chloride penetration test.                                                                           | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 8.     | Flow table motorized  | To measure the consistency and workability of cement and concrete. It helps determine the flow characteristics of the material by subjecting it to a series of mechanical movements. | ✓ To assess the workability, consistency, and ease of handling of fresh cement and concrete mixtures.                                                            | For Project & Consultancy work | The motorized flow table adheres to standard guidelines for flowability testing, ensuring that students can work within the recognized testing parameters used in the construction industry | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 9.     | Vee bee Consistometer | Device used to measure the workability or consistency of concrete, specifically for mixes that are too stiff for traditional slump testing                                           | ✓ By measuring how concrete behaves under vibration, it helps determine how easily it can be placed and compacted, ensuring the                                  | For Project & Consultancy work | By developing a Vee-Bee Consistometer, students gain insight into both the theoretical and practical aspects of concrete workability testing,                                               | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11,                              |



| Sl. No | Facility Name               | Details                                                                                                                                                                                                                                                                                                             | Reason(s) for creating facility                                                                                                                                                                                                                             | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                  | Relevance to POs/PSOs                                                            |
|--------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                             |                                                                                                                                                                                                                                                                                                                     | quality and performance of the final concrete product.                                                                                                                                                                                                      |                                | contributing to a well-rounded understanding of material testing and quality control in construction                                                                                                             | PO12, PSO1, PSO2, PSO3, PSO4                                                     |
| 10.    | Blains Apparatus            | Determining the specific surface area of materials, particularly fine powders or cement. It is commonly used in the study of materials that have very small particles, such as those in cement, which require measurement of their surface area for various quality control and material characterization purposes. | To assess the fineness of cement. Finer particles in cement result in a higher surface area, leading to a faster setting time and greater strength development.                                                                                             | For Project & Consultancy work | Blain's apparatus would allow students to gain practical experience in designing scientific equipment and understanding the underlying principles of material science, fluid dynamics, and laboratory techniques | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 11.    | Compressometre              | To measure the deformation or strain in the longitudinal direction (along the length) of a material or specimen when it is subjected to compressive forces.                                                                                                                                                         | It is commonly used in materials testing to assess how a material behaves under compression, which is particularly important for understanding the elastic properties and structural integrity of materials like concrete, metal, polymers, and composites. | For Project & Consultancy work | Students use a longitudinal compressometer, they are engaging with an essential tool for understanding how materials behave under compressive stress.                                                            | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 12.    | Compaction factor apparatus | To measure the workability of fresh concrete by determining its compaction factor. It is especially used in cases where the concrete mixture is too stiff to                                                                                                                                                        | ✓ TO determine the Workability of the concrete                                                                                                                                                                                                              | For Project Work               | This test is particularly useful for mixes where the concrete's consistency cannot be easily determined using traditional slump tests, such as high-density,                                                     | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12,                        |

| Sl. No | Facility Name                  | Details                                                                                                                                                                                                         | Reason(s) for creating facility                        | Utilization            | Areas in which students' are expected to have enhanced learning                                                                                                                                                  | Relevance to POs/PSOs                                                            |
|--------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                                | measure workability using the slump test.                                                                                                                                                                       |                                                        |                        | low-slump concrete mixtures.                                                                                                                                                                                     | PSO1, PSO2, PSO3, PSO4                                                           |
| 13.    | Flash & Fire Point             | To evaluate the flammability and fire hazards of liquids, especially in fuels, oils, and chemicals. These two properties help determine the temperature at which a substance can ignite and sustain combustion. | ✓ To determine the flammability hazards of the bitumen | For Consultancy work   | Pensky-Martens Flash Point Tester (Closed Cup Method): The sample is heated in a closed container, and the temperature at which a spark causes a flash is recorded                                               | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 14.    | Mechanical Extensometer        | To measure the elongation (strain) or deformation of a material when subjected to mechanical stress.                                                                                                            | ✓ To determine the Elongation of th Concrete           | For Consultancy work   | A mechanical extensometer is an essential tool for measuring the strain (elongation or deformation) of materials under load. It is especially useful in tensile testing and other material property evaluations. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 15.    | Benkelman beam with dial gauge | To measure the deflection of road pavements under a moving wheel load.                                                                                                                                          | ✓ To determine the Deformation of the Road Pavement    | Academic & Consultancy | The Benkelman Beam is an essential tool for studying the deflection of flexible pavements under load, and students can learn a lot about road design, maintenance, and material behavior through its application | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2,            |

| Sl. No                                                                                    | Facility Name           | Details                                                                                                                                                  | Reason(s) for creating facility                                                                                                                                                                    | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                                     | Relevance to POs/PSOs                                                            |
|-------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                                                                                           |                         |                                                                                                                                                          |                                                                                                                                                                                                    |                                |                                                                                                                                                                                                                                     | PSO3, PSO4                                                                       |
| 16.                                                                                       | Concrete Impact machine | To assess the impact resistance or toughness of concrete materials                                                                                       | ✓ To Determine the Impact Resistance of the Concrete Disk                                                                                                                                          | Project & Consultancy work     | It measures how well concrete can withstand sudden, short-duration forces, which is essential for determining the material's performance in environments subjected to shock loads, such as roads, pavements, and industrial floors. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 17.                                                                                       | Coating thickness gauge | To measure the thickness of coatings (such as paint, plating, or other protective layers) on a substrate material (such as metal, plastic, or concrete). | ✓ To determine the surface Tension behaviour of the material                                                                                                                                       | For Project & Consultancy work | Students can accurate and non-destructive way to measure coating thickness and ensure that materials meet performance and safety standards.                                                                                         | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| <b>3. ENVIRONMENTAL ENGINEERING LABORATORY / WATER AND WASTEWATER ANALYSIS LABORATORY</b> |                         |                                                                                                                                                          |                                                                                                                                                                                                    |                                |                                                                                                                                                                                                                                     |                                                                                  |
| 18.                                                                                       | Spectrophotometer       | To determine the concentration of various compounds in water and wastewater sample like phosphate, ammonia nitrogen, fluorides etc                       | ✓ To understand the correlation about the concentration of pollutants and its light absorption and transmittance capacity.<br>✓ To estimate the unknown sample concentration by comparing standard | For Project & Consultancy work | Characterization of chemical parameters of water and wastewater samples.                                                                                                                                                            | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2,            |

| Sl. No | Facility Name    | Details                                                                                                                                                                                                                                                                                                                             | Reason(s) for creating facility                                                             | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                                                                                                                                                                                    | Relevance to POs/PSOs                                                           |
|--------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|        |                  |                                                                                                                                                                                                                                                                                                                                     | solutions using Beer-Lambert law.                                                           |                                |                                                                                                                                                                                                                                                                                                                                                                                    | PSO3, PSO4                                                                      |
| 19.    | Flame Photometer | An analytical instrument used to measure the concentration of certain metal ions, such as sodium (Na), potassium (K), calcium (Ca).It works on the principle of flame emission spectroscopy, where the sample is introduced into a flame, and the emitted light from the excited ions is measured to determine their concentration. | ✓ Measure elements like sodium, potassium, calcium                                          | For Project & Consultancy work | The application of flame photometry in their studies helps develop both technical skills and theoretical knowledge, and fosters a deeper understanding of scientific principles.                                                                                                                                                                                                   | PO1, PO2, PO3, PO4, PO5, PO6, PO7,PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 20.    | Desiccator       | A desiccator is a sealed container used in laboratories to store moisture-sensitive materials in a dry environment.                                                                                                                                                                                                                 | ✓ Preserving, drying, and storing moisture-sensitive materials in a controlled environment. | For Project & Consultancy work | Students gain hands-on experience with essential scientific tools and techniques, fostering deeper engagement with topics like chemistry, biology, and material science. The use of desiccators helps develop critical lab skills, enhances understanding of environmental control in experiments, and encourages the application of theoretical knowledge to real-world problems. | PO1, PO2, PO3, PO4, PO5, PO6, PO7,PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |



| Sl. No | Facility Name          | Details                                                                                                                                                                                                                                                                                                       | Reason(s) for creating facility                                                                                                                      | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                                                                                                                                  | Relevance to POs/PSOs                                                            |
|--------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 21.    | Water Bath Rectangular | It consists of a container filled with heated water in which smaller containers (such as test tubes, flasks, or beakers) can be submerged. The water provides a uniform, controlled temperature environment, ensuring gentle and even heating                                                                 | ✓ Heating substances or maintaining a stable temperature over an extended period.                                                                    | For Project & Consultancy work | Students learn how to apply theoretical knowledge in practical contexts, enhancing their understanding of temperature-dependent processes, reactions, and biological systems.                                                                                                                                                    | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 22.    | Distillation Unit      | A system designed to purify water by separating impurities, salts, and other contaminants through the process of distillation. In this process, water is boiled to produce steam, which then condenses back into liquid water, leaving impurities behind. The purified water is collected as distilled water. | ✓ Distilled water is used for mixing reagents, preparing solutions, and rinsing glassware to avoid any residues or ions that could affect reactions. | For Project & Consultancy work | Using a distillation unit enhances students' learning across a variety of disciplines, including chemistry, biology, environmental science, and engineering. They develop practical laboratory skills, enhance their understanding of fundamental scientific principles, and learn to apply these skills to real-world problems. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 23.    | Filtration Assembly    | A system designed to remove impurities, particles, and contaminants from liquids or gases by passing them through a filter medium.                                                                                                                                                                            | ✓ Filtration is widely used in various applications, such as water purification, air filtration, chemical processing, and laboratory work.           | For Project & Consultancy work | By using a Filtration Assembly, students enhance their learning in chemical separation techniques, laboratory skills, environmental science, and industrial applications. They develop practical, technical skills in                                                                                                            | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2,            |

| Sl. No | Facility Name               | Details                                                                                                                                                                                         | Reason(s) for creating facility                                                                                                                                                                                                | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                                                                                                                                                                               | Relevance to POs/PSOs                                                            |
|--------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                             |                                                                                                                                                                                                 |                                                                                                                                                                                                                                |                                | handling and optimizing filtration processes, making this fundamental technique essential for students pursuing careers in chemistry, biology, environmental science, and engineering.                                                                                                                                                                                        | PSO3, PSO4                                                                       |
| 24.    | Spectroquant Thermo reactor | A device used in laboratory settings to facilitate various types of chemical analyses, particularly for water quality testing,                                                                  | ✓ Spectroquant Thermo Reactor is in the determination of COD in water. COD is a measure of the organic matter in water and is a critical parameter for assessing water quality, especially in wastewater treatment facilities. | For Project & Consultancy work | Using a Spectroquant Thermo Reactor in laboratory settings provides students with a wealth of opportunities to enhance their learning in areas such as analytical chemistry, environmental science, water quality analysis, and instrumental techniques. They develop valuable skills in data interpretation, instrumentation, quality control, and environmental monitoring. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 25.    | Gas Chromatography          | Gas Chromatography works on the principle of separation based on the interaction of compounds with a stationary phase and the flow of a mobile phase (usually a gas, often helium or nitrogen). | ✓ Gas Chromatography (GC) is a widely used analytical technique for separating, identifying, and quantifying volatile and semi-volatile compounds in a mixture.                                                                | For Project & Consultancy work | The use of Gas Chromatography (GC) in student laboratories enhances learning in a wide array of scientific fields, ranging from analytical chemistry to environmental science, forensic science, and                                                                                                                                                                          | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2,            |

| Sl. No | Facility Name              | Details                                                                                                                                                                                                                                                               | Reason(s) for creating facility                                                                                                               | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                                                                      | Relevance to POs/PSOs                                                            |
|--------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                            |                                                                                                                                                                                                                                                                       |                                                                                                                                               |                                | food chemistry. Students acquire hands-on experience with advanced instrumentation, chemical separations, and quantitative analysis, gaining practical skills that are widely applicable in research, industry, and environmental monitoring.                        | PSO3, PSO4                                                                       |
| 26.    | Kjeldhal Digestion Unit    | The Kjeldahl Digestion Unit is an essential laboratory apparatus used in the Kjeldahl method, which is designed to determine the nitrogen content in organic and inorganic samples.                                                                                   | ✓ Measures nitrogen in wastewater, sludge, and sediments to monitor pollution levels.                                                         | For Project & Consultancy work | The Kjeldahl method is foundational in various scientific fields, and mastering it enhances students' practical knowledge and critical thinking, preparing them for advanced study and careers in research, agriculture, environmental science, and food industries. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 27.    | Kjeldhal Distillation Unit | The Kjeldahl Distillation Unit is a critical component of the Kjeldahl method, used to quantify nitrogen in a sample after the digestion step. This unit focuses on converting ammonium ions (produced during digestion) into ammonia gas, which is then captured and | ✓ Quantifies nitrogen compounds in water bodies, wastewater, and sludge, aiding in pollution monitoring and treatment efficiency evaluations. | For Project & Consultancy work | The use of a Kjeldahl Distillation Unit enhances student learning in a variety of scientific fields, from analytical chemistry and biochemistry to environmental science and agriculture. Students gain practical experience in nitrogen analysis,                   | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl. No | Facility Name          | Details                                                                                                                                                         | Reason(s) for creating facility                                                                                                    | Utilization                    | Areas in which students' are expected to have enhanced learning                                                                                                                                                                                                                                    | Relevance to POs/PSOs                                                            |
|--------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                        | measured for nitrogen content determination.                                                                                                                    |                                                                                                                                    |                                | distillation, titration, and experimental design, while deepening their understanding of nitrogen's role in biological systems, soil health, and food chemistry.                                                                                                                                   |                                                                                  |
| 28.    | Ion 6+PH/Ion/ORP Meter | The Ion 6+ pH/Ion/ORP Meter is a versatile analytical device used to measure pH, ion concentration, and oxidation-reduction potential (ORP) in various samples. | ✓ Monitoring pH and ORP to control treatment processes. Measuring ion concentrations (e.g., fluoride, chloride) in water supplies. | For Project & Consultancy work | The use of an Ion 6+ pH/Ion/ORP Meter enhances students' understanding across a broad range of scientific disciplines, including chemistry, environmental science, biology, and industrial applications. By learning how to measure pH, ion concentration, and ORP, students gain practical skills | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 29.    | Chlorine Comparator    | A Chlorine Comparator is a portable device used to measure the concentration of chlorine in water.                                                              | ✓ Ensures proper chlorination in drinking water and wastewater systems.                                                            | For Project & Consultancy work | The use of a chlorine comparator in laboratory exercises helps students develop a range of skills in chemical analysis, data interpretation, and scientific problem-solving.                                                                                                                       | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl. No                              | Facility Name                                                           | Details                                                                                                                                                 | Reason(s) for creating facility                                                                                                                              | Utilization                      | Areas in which students' are expected to have enhanced learning                                                                                                                           | Relevance to POs/PSOs                                                                 |
|-------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 30.                                 | Compound Microscope                                                     | An essential tool in biology, medicine, materials science, and education.                                                                               | ✓ Observing cells, tissues, microorganisms, and pathological samples.                                                                                        | For Project & Consultancy work   | A compound microscope enhances student learning in a variety of scientific areas, from practical laboratory techniques to theoretical knowledge in optics, biology, and material science. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4      |
| <b>4. SOIL MECHANICS LABORATORY</b> |                                                                         |                                                                                                                                                         |                                                                                                                                                              |                                  |                                                                                                                                                                                           |                                                                                       |
| 31.                                 | Standard penetration test Apparatus                                     | To measure the resistance of the soil strata to the penetration undergone.<br><br>To provide information on geotechnical engineering properties of soil | ✓ To understand the correlation between the soil properties and the penetration resistance.<br><br>✓ To determine the Actual soil behaviour using SPT values | For Consultancy Work             | Foundation Engineering - Acquire knowledge in selection of suitable foundation method                                                                                                     | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 32.                                 | Dry Sieve Test (Sieve Shaker)                                           | Mechanical Shaker                                                                                                                                       | ✓ To segregate the soil mass under different sizes (Soil Classification)                                                                                     | Academic & Consultancy           | Geotechnical Engineering : Index Properties of Soils                                                                                                                                      | Dry Sieve Test (Sieve Shaker)                                                         |
| 33.                                 | Interface Friction measurement Apparatus for modified direct shear test | Mechanical device Specially designed to evaluate both 'frictional' as well as 'pull out resistance' of soil reinforcing                                 | ✓ To find the pullout resistance of other materials                                                                                                          | Academic, Research & Consultancy | Soil Reinforcements : Mechanically Stabilized Walls                                                                                                                                       | Interface Friction measurement Apparatus for modified direct shear test               |



| Sl. No | Facility Name                     | Details                                                                                                   | Reason(s) for creating facility                     | Utilization                      | Areas in which students' are expected to have enhanced learning | Relevance to POs/PSOs             |
|--------|-----------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------|-----------------------------------------------------------------|-----------------------------------|
| 34.    | Geotextile Permeameter            | Mechanical device<br>Specially designed to test the requirements of water permeability of coir geotextile | ✓ To find the water permeability of coir geotextile | Academic, Research & Consultancy | Soil Reinforcements : Mechanically Stabilized Walls             | Geotextile Permeameter            |
| 35.    | Hydrodynamic Sieve Test Apparatus | Mechanical device                                                                                         | ✓ For determining the porometry of the geotextile   | Academic, Research & Consultancy | Soil Reinforcements : Mechanically Stabilized Walls             | Hydrodynamic Sieve Test Apparatus |

#### 5. SURVEYING LABORATORY

|     |                        |                                                                                                                                                                                                    |                                                                                                                               |                                         |                                                                                                                                                                        |                                                                                  |
|-----|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 36. | ✓ GPS                  | ✓ <b>Model:</b> Garmin Etrex - 10.<br><b>Memory:</b> 1000 points. The etrex® 10 allows for the transfer of waypoint and track data between the GPS and the computer.                               | ✓ To find out the Latitude and Longitude of the points.<br>✓ To do the projects work in the field of surveying.               | During Survey camp                      | <b>Surveying</b> – To enhance the technical skill in field of survey and to obtain knowledge in handling the advanced equipment available in the Surveying Laboratory. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 37. | ✓ Laser distance meter | ✓ Measuring Range: 0.05m – 150m both indoors and outdoors<br>✓ Measuring Accuracy $\pm 1$ mm<br>✓ Digital Pointfinder / Viewfinder with 4x zoom.<br>✓ Measuring Units m, in, ft & 20 Points Memory | ✓ With the help of LDM, the students will find out the area, volume, distance and perimeter of the points in a simple manner. | For affiliation and accreditation works |                                                                                                                                                                        |                                                                                  |
| 38. | ✓ Total Station        | ✓ <b>Model:</b> Sokkia.2LS CygnusKS -102<br>✓ <b>Accuracy:</b> 2"<br><b>Least count:</b> 1"<br><b>Measuring Distance (approx.):</b> 200m & Reflectorless mode:                                     | ✓ Foundation Marking, Levelling Work, and Contouring work carried out for our college, schools and consultancy works.         | For establishment of playground         |                                                                                                                                                                        |                                                                                  |

| Sl. No                                                                             | Facility Name                             | Details                                                                  | Reason(s) for creating facility                                                                                | Utilization                                               | Areas in which students' are expected to have enhanced learning                                        | Relevance to POs/PSOs                                                            |
|------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                                                                                    |                                           | 2Km. (with single prism).<br>✓ Int. memory: 24000 pts.                   |                                                                                                                |                                                           |                                                                                                        |                                                                                  |
| <b>6. Centre for Geospatial Technology</b>                                         |                                           |                                                                          |                                                                                                                |                                                           |                                                                                                        |                                                                                  |
| 39.                                                                                | ✓ Differential Global Positioning system, | ✓ <b>Model: Geomate SG7 Handheld GNSS Data Collector</b>                 | ✓ Students need to Kown Recent survey system and How to survey the land with less time<br>✓ Consultancy work   | VAC & Consultancy work                                    | Used in mapping, land surveying, and infrastructure planning, where accurate spatial data is critical. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 40.                                                                                | ✓ Drone                                   | ✓ Avio-M1 Professional Mapping Drone with PPK system,                    | ✓ Students need to Kown Recent survey system and How to survey the land with less time<br>✓ Consultancy work   | VAC & Consultancy work                                    | Used in mapping, land surveying, and infrastructure planning, where accurate spatial data is critical. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| <b>FLUID MECHANICS &amp; MACHINERY LABORATORY/HYDRAULIC ENGINEERING LABORATORY</b> |                                           |                                                                          |                                                                                                                |                                                           |                                                                                                        |                                                                                  |
| 41.                                                                                | ✓ Impact of Jet on Vanes                  | ✓ To study the force and momentum change when a fluid is striking a vane | ✓ The hydrodynamic force created will enable the graduates to have better understanding on working of turbines | ✓ To get Practical Exposure in Water Resource Engineering | ✓ Turbomachinery studies                                                                               | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2,            |

| Sl. No | Facility Name                           | Details                                                   | Reason(s) for creating facility                                          | Utilization                                               | Areas in which students' are expected to have enhanced learning | Relevance to POs/PSOs                                                            |
|--------|-----------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                                         |                                                           |                                                                          |                                                           |                                                                 | PSO3, PSO4                                                                       |
| 42.    | ✓ Pitot Tube Setup                      | ✓ To measure the local fluid flow velocity                | ✓ To have better understanding flow rate measurements                    | ✓ To get Practical Exposure in Water Resource Engineering | ✓ Discharge measuring devices                                   | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 43.    | ✓ Multi-Stage Centrifugal Pump Test Rig | ✓ To increase the fluid pressure and to produce high head | ✓ To enhance the knowledge on various types and stages of pumps          | ✓ To get Practical Exposure in Water Resource Engineering | ✓ Turbomachinery studies                                        | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 44.    | ✓ Venturiflume Apparatus                | ✓ To measure the flow rate in rivers streams              | ✓ To have better understanding venturi effect and flow rate measurements | ✓ To get Practical Exposure in Water Resource Engineering | ✓ Discharge measuring devices                                   | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| <b>Sl. No</b> | <b>Facility Name</b>                           | <b>Details</b>                                                      | <b>Reason(s) for creating facility</b>                                     | <b>Utilization</b>                                        | <b>Areas in which students' are expected to have enhanced learning</b> | <b>Relevance to POs/PSOs</b>                                                     |
|---------------|------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 45.           | ✓ Kaplan Turbine Test Rig (Electrical Loading) | ✓ To determine the characteristic curves and efficiency of turbines | ✓ Develop knowledge on conversion of hydraulic energy to electrical energy | ✓ To get Practical Exposure in Water Resource Engineering | ✓ Turbomachinery studies                                               | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

## **6.3 Laboratories: Maintenance and Overall Ambiance: (10)**

### **6.3.1 Purchase process**

All the laboratories in the department are equipped with state of the art instruments. The department follows a well-structured consumables/equipment purchase system. Fig.6.3.1 shows the equipment purchase process followed. (Link : [https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil\\_NBA\\_Criteria\\_VI.pdf](https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil_NBA_Criteria_VI.pdf))

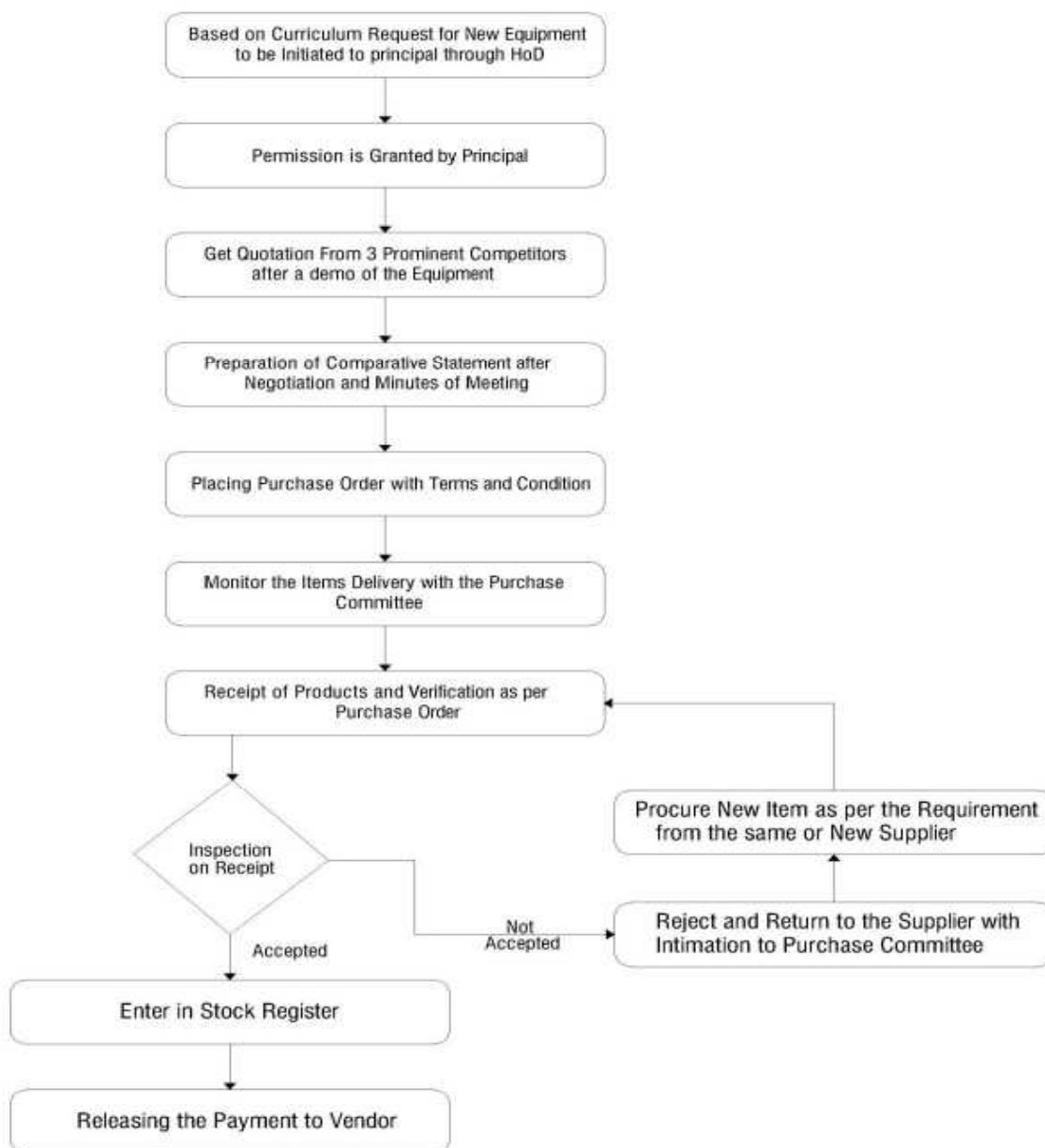


Fig 6.3.1 Equipment Purchase Process

### **6.3.2 Maintenance of Laboratory Equipment**

The Department of Civil engineering is maintained daily with sufficient numbers of housekeeping persons other additional task are summarized below

- All the laboratories are equipped with continuous power supply ensuring uninterrupted operation of machines.
- One teaching faculty and minimum of one laboratory instructor take care of the overall functioning / maintenance of the laboratory.
- Stock registers are maintained in the laboratory for capital goods and consumables
- Separate maintenance records are available for each and every laboratory where maintenance activities are recorded.
- Cleaning of machine and equipment is carried out by the lab assistant and it is recorded in separate Equipment maintenance register.
- Cleaning of floors, doors, windows, carpentry work, and painting work are carried out by the Housekeeping and Estate Office, and the details are recorded in the General Maintenance Register
- Regular checkup of equipment / computer is carried out as and when required.
- Weekly oil level check-up is done in certain laboratories by the lab assistant.
- Major repairs of machines are carried out by authorized agencies or service engineers, and the details of the work are recorded in the Service Register.
- The utilization of laboratories and machinery/equipment is recorded in a separate Utilization /Student Login Register.
- Licensed Software is updated whenever required and all the latest software is uploaded to meet the industrial needs in certain labs.
- All computers in CAD lab are uploaded with latest analysis and design software with correlation with Bentley Systems to meet the Industrials Needs.



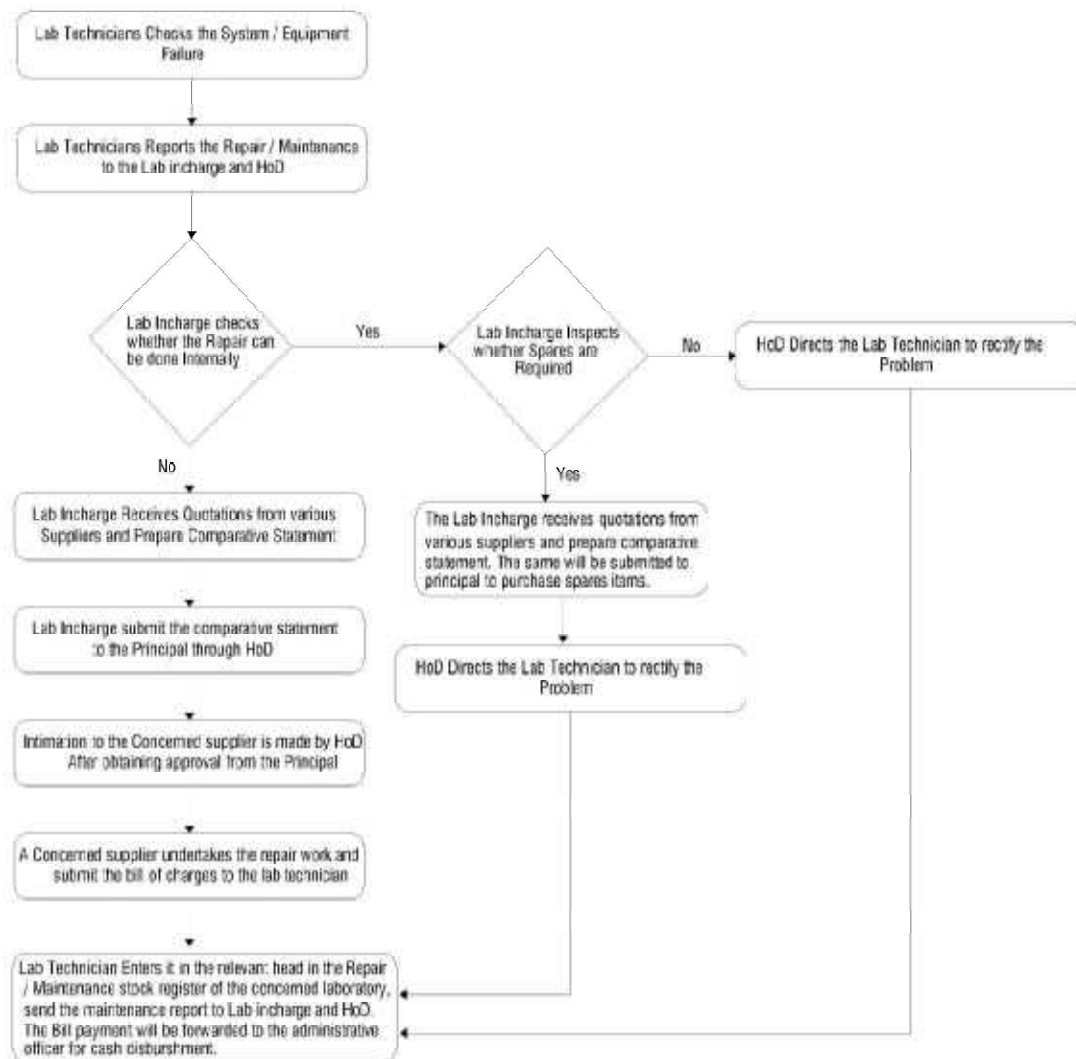


Fig 6.3.2 Steps Involved in Maintenance and Repair Work

(Link : [https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil\\_NBA\\_Criteria\\_VI.pdf](https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil_NBA_Criteria_VI.pdf))



Fig.6.3.3 Sample Maintenance record for Concrete laboratory

(Link : [https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil\\_NBA\\_Criteria\\_VI.pdf](https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil_NBA_Criteria_VI.pdf))

### **6.3.3 Overall Ambience**

#### **CAD – DIGITALIZED CONSTRUCTION LABORATORY**

- CAD laboratories are equipped with fans, good ventilation and ambience
- Cad laboratory have adequate number of furniture's, ergonomics benches/ chairs.
- Preventive maintenance are carried out for all system hardware and software
- Maintain the sufficient hardware and licensed versions of software in the laboratory.
- Edibles are strictly prohibited inside the laboratory.
- Display Boards are provided for displaying the course syllabus.
- Good quality window glasses with curtains are provided to avoid direct exposure of sunlight inside the laboratory.
- Provided with sufficient lighting facilities for better luminance.
- Provided individual MCB for handling emergency situations.
- Vision, Mission, PEO's, PO's, signboard display for better understanding
- Equipped with rolling chair and good finish floor tiles for better appearance.
- Movable sliding keyboard drawer are provided for ease access of hardware of the computer.
- Wooden furniture's foot holders are provided to rest the foot during the system operation. Servicing of each laboratory is doing frequently.
- Each Laboratory are equipped with First Aid Box and Fire Extinguisher.
- Each Lab is equipped with white/black board.

#### **Concrete And Highway Engineering Laboratory, Soil Mechanics Laboratory:**

- Experienced and trained technicians are employed in the laboratory for carry out maintenance work.
- Excess spare parts, consumables are procured for immediate replacement, if it is needed, adequate inventories are maintained in the laboratory to address the maintenance issue.
- After machining, the machines are cleaned and derbies are removed.
- Oil lubrication level, water level is checked regularly, Adequate oil are procured and stock should be maintained before commencement of laboratory classes in the semester.
- The machines are arranged with prescribed layout for easy access.
- The machines specifications are provided for easy identification of Machines.
- Measuring equipment are calibrated with standard. Servicing of each laboratory is doing frequently.
- Each Laboratories are equipped with First Aid Box and Fire Extinguisher.
- Each Lab is equipped with white/black board.

#### **Strength of Materials Laboratory, Fluid Mechanics & Machinery Laboratory/Hydraulic Engineering Laboratory:**

- Machines are arranged in prescribed layout to easily exercise machines.
- Instruments are calibrated and necessary trainings or given to technicians to calibrate the equipment, if needed.

- Issues, breakdown of engines are immediately addressed with technicians or with experts from engine suppliers end, if needed
- Enough ventilation is provided for enough air circulation inside the laboratory and removing exhaust gases.
- Ensure all the pipelines are leak proof and use proper gaskets for arresting leakage.
- Maintain water pressure within safe range.
- Properly clean the equipment for preventing first and periodically clean the water storage tank to avoid corrosion.
- Oil lubrication level, water level is checked regularly, Adequate oil are procured and stock should be maintained before commencement of laboratory classes in the semester.
- Servicing of each laboratory is doing frequently.
- Each Laboratories are equipped with First Aid Box and Fire Extinguisher.
- Each Lab is equipped with white/black board.

### **Surveying Laboratory:**

- Experienced and trained technicians are employed in the laboratory for carry out maintenance work.
- Excess spare parts, consumables are procured for immediate replacement, if it is needed, adequate inventories are maintained in the laboratory to address the maintenance issue.
- The survey laboratory provides with ample space for equipment, workstations, and movement. This ensures ease of access to tools and instruments, as well as the ability to conduct experiments without feeling cramped.
- The laboratory situated on stable floor, away from sources of vibrations or disturbances that could affect instruments and measurements.
- Proper storage facilities for equipment, tools, and materials contribute to an efficient and organized laboratory environment. This includes designated areas for storing survey instruments, reference materials, etc.,
- The laboratory equipped with the necessary technical infrastructure, including computers, software for data analysis, and internet connectivity for research and collaboration.
- Enough ventilation is provided for enough air circulation inside the laboratory and removing exhaust gases.
- Servicing of each laboratory is doing frequently.
- Each Laboratories are equipped with First Aid Box and Fire Extinguisher.
- Each Lab is equipped with white/black board.

### **Environmental Engineering Laboratory/Water and Waste Water Analysis Laboratory**

- The laboratory equipped with appropriate safety equipment such as eyewash stations, safety showers, and fire extinguishers. Additionally, personnel are trained in proper safety protocols.
- Provided with A well-designed ventilation system for maintaining air quality and removing hazardous fumes and odors generated during testing and analysis. Adequate ventilation helps protect personnel and ensures accurate results.
- Provided with sufficient lighting facilities for better luminance.
- Excess spare parts, consumables are procured for immediate replacement, if it is needed, adequate inventories are maintained in the laboratory to address the maintenance issue.
- Instruments are calibrated and necessary trainings or given to technicians to calibrate the equipment, if needed.

- Servicing of each laboratory is doing frequently.
- Each Laboratories are equipped with First Aid Box and Fire Extinguisher.
- Each Lab is equipped with white/black board.

**Table 6.1 List of Classrooms and Laboratories**

| <b>Sl. No.</b> | <b>Name of the Laboratory</b>               | <b>Location</b> | <b>Lab No.</b> | <b>Area (Sq.m)</b> |
|----------------|---------------------------------------------|-----------------|----------------|--------------------|
| 1.             | Concrete and Highway Engineering Laboratory | Ground Floor    | C0R04          | 140                |
| 2.             | Fluid Mechanics and Machinery Laboratory    | Ground Floor    | C0L01          | 181                |
| 3.             | Surveying Laboratory                        | First Floor     | C1L01          | 165                |
| 4.             | Water and Wastewater Analysis Laboratory    | First Floor     | C1L05          | 109                |
| 5.             | Soil Mechanics Laboratory                   | First Floor     | C1L04          | 148                |
| 6.             | CAD – Digitalized Construction Laboratory   | First Floor     | C1L01-B        | 110                |
| 7.             | Strength of Materials Laboratory            | Ground Floor    | C0L03          | 124                |
| 8.             | Construction Practice Lab                   | Ground Floor    | CPL-01         | 66                 |
| 9.             | Centre for Geospatial Technology            | First Floor     | C1L01-A        | 68                 |
| 10.            | HOD Room & Department                       | First Floor     | C1L02          | 122                |
| 11.            | Project Laboratory                          | Ground Floor    | C0R06          | 23.6               |
| 12.            | Department Library                          | First Floor     | C1L06          | 57.6               |
| 13.            | Seminar Hall                                | Second Floor    | B2L04          | 152                |
| 14.            | Class Room -I                               | First Floor     | B1L02          | 82                 |
| 15.            | Class Room – II                             | Third Floor     | B3L05          | 82                 |
| 16.            | Class Room – III                            | Third Floor     | B3L03          | 82                 |
| 17.            | Class Room – IV                             | Third Floor     | B3L04          | 82                 |
| 18.            | Research Centre                             | First Floor     | C1L09          | 21.93              |

## 6.4 Project laboratory: (5)

### 6.4.1 Facilities provided for projects

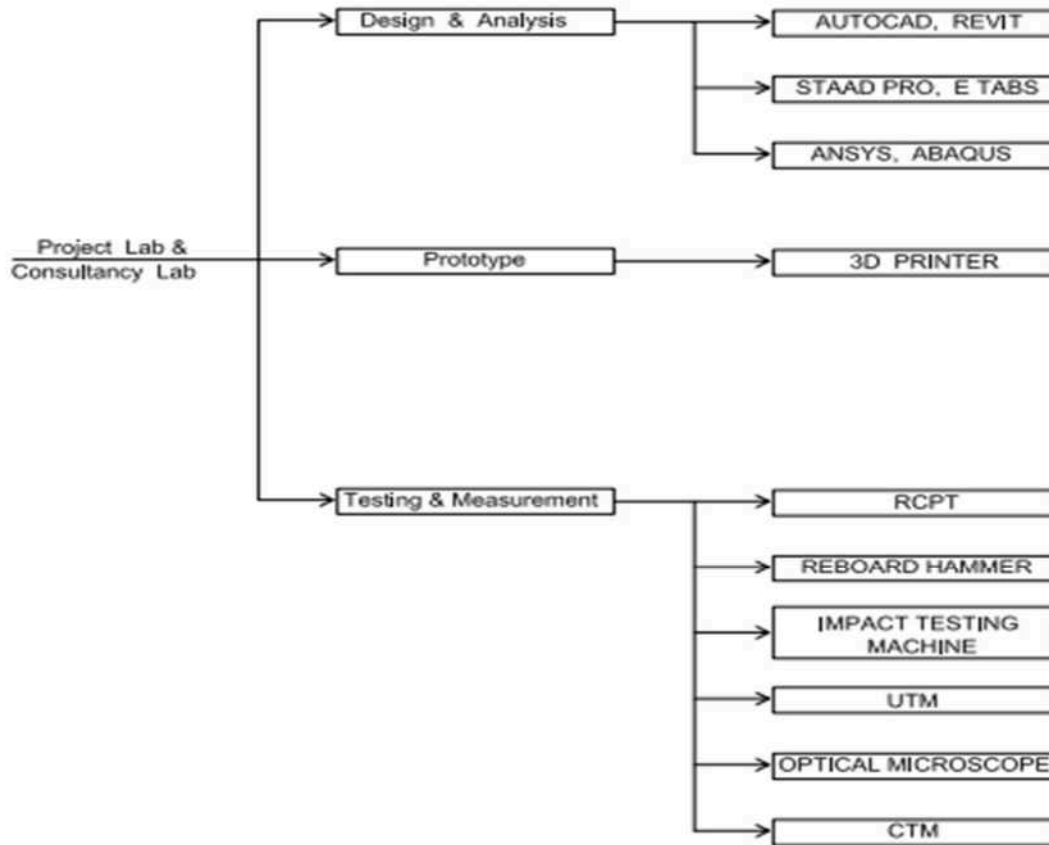


Fig.6.4.1 Facilities provided for projects

(Link : [https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil\\_NBA\\_Criteria\\_VI.pdf](https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil_NBA_Criteria_VI.pdf))

| Sl. No. | Facilities                                        | Utilization                                             |
|---------|---------------------------------------------------|---------------------------------------------------------|
| 1.      | <b>Project Laboratory</b>                         | Utilized for Student projects and Faculty Research work |
|         | • Rapid Chloride Penetration Test equipment       |                                                         |
|         | • Concrete Impact testing                         |                                                         |
|         | • Rebound Hammer                                  |                                                         |
|         | • Standard Penetration test equipment             |                                                         |
|         | • Rapid Moisture Meter                            |                                                         |
| 2.      | <b>COMPUTER AIDED BUILDING DRAWING LABORATORY</b> | Utilized for Student projects and Faculty Research work |
|         | • AutoCAD (2023),                                 |                                                         |



| Sl. No. | Facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Utilization                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
|         | <ul style="list-style-type: none"> <li>• STAAD.Pro V8i,</li> <li>• Revit Architecture 2016</li> <li>• ETABS 17</li> <li>• Primavera Contractor P6</li> <li>• ANSYS 17.0</li> <li>• Abaqus</li> </ul>                                                                                                                                                                                                                                                                                                                        |                                                         |
| 3.      | <b>CONCRETE AND HIGHWAY ENGINEERING LABORATORY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                         |
|         | <ul style="list-style-type: none"> <li>• Concrete impact testing machine</li> <li>• SCC testing equipments (L box, U box, V- funnel, J Ring &amp; Flow table)</li> <li>• CTM 2000 KN capacity</li> <li>• Los angles abrasion testing machine</li> <li>• Universal penetrometer</li> <li>• Marshall stability</li> <li>• Benkelman beam</li> <li>• Bitumen centrifuge extractor</li> <li>• Standard tar viscometer</li> <li>• Flash &amp; fire point</li> <li>• Ring and ball apparatus</li> <li>• Sieve Analysis</li> </ul> | Utilized for Student projects and Faculty Research work |
| 4.      | <b>WATER AND WASTEWATER ANALYSIS LABORATORY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                         |
|         | <ul style="list-style-type: none"> <li>• Oxygen Analyzer</li> <li>• Spectrophotometer</li> <li>• Ion selective Electrode</li> <li>• Flame Photometer</li> <li>• Gas Chromatography</li> <li>• Digital Turbidity Meter</li> <li>• BOD Analyzer, COD Analyzer</li> <li>• Muffle Furnace, pH Meter</li> <li>• Bacteriological Incubator</li> <li>• Conductivity Meter</li> </ul>                                                                                                                                               | Utilized for Student projects and Faculty Research work |
| 5.      | <b>Consultancy Laboratory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                         |
|         | <ul style="list-style-type: none"> <li>• Total Station</li> <li>• Line Laser Level</li> <li>• GPS</li> <li>• DGPS</li> <li>• Compression Testing Machine</li> <li>• Spectrophotometer</li> <li>• Ion selective Electrode</li> </ul>                                                                                                                                                                                                                                                                                         | Utilized for Consultancy Works                          |

| Sl. No. | Facilities                                                                   | Utilization |
|---------|------------------------------------------------------------------------------|-------------|
|         | <ul style="list-style-type: none"> <li>Flame Photometer</li> </ul>           |             |
|         | <ul style="list-style-type: none"> <li>BOD Analyzer, COD Analyzer</li> </ul> |             |
|         | <ul style="list-style-type: none"> <li>pH Meter</li> </ul>                   |             |

### 6.4.2 Utilization and outcomes

Project works are being carried out in individual laboratories and industries by students. Also, a separate project laboratory of size 23.6m with library for reference, systems and internet facilities to process the results/data, Laboratories can offer students and faculty valuable opportunities for projects beyond traditional academic work, fostering innovation and skill-building in practical, hands-on settings. By utilizing the lab space and resources, students can engage in research projects, independent study, or extracurricular projects that deepen their knowledge and technical skills. Faculty can also benefit by using the lab for professional development or to collaborate on interdisciplinary projects with colleagues and students, often leading to publications, prototypes, or innovative solutions to real-world problems. Additionally, labs can host workshops, seminars, or community engagement events, providing a space to experiment with new ideas, mentor students, and build networks. This broader use of lab resources not only enriches the academic environment but also promotes a culture of curiosity, creativity, and continuous learning.

**\*\*Facilities Provided:**

The Following Facilities are created for doing projects in this laboratory

#### STUDENTS PROJECTS WITH OUTCOMES (2021 – 2024)

| Sl.No. | Team Members                                                     | Details of the Project    | Facilities Utilized             | Outcome of the projects                                                                                                                                                                                                          | Achievements of PO/PSO                                                                |
|--------|------------------------------------------------------------------|---------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1.     | Balaji K<br>Devar Piran M<br>Arun Kumar S<br>Deena<br>Dhayalan R | Household Waste Composter | pH meter,<br>Conductivity Meter | Won Cash prize worth Rs.5000 in the event of Current Environmental Challenges and solution towards sustainability, EUPHORIA'24 organized by the Department of Civil Engineering, Kalasalingam Academy of Research and Education. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl.No. | Team Members      | Details of the Project                                                     | Facilities Utilized                                        | Outcome of the projects                                                                                                                                                              | Achievements of PO/PSO                                                                |
|--------|-------------------|----------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 2.     | A. Keerthi roja   | Impact and Mechanical Properties of carbon fibre reinforced Concrete.      | CTM (2000 kN capacity), Impact Testing Machine             | Leema Margret, A. & Keerthi, Roja (2024). Effect of Addition of Carbon Fibre on Mechanical Properties of Concrete. Journal of Science Technology and Research (JSTAR) 5 (1):535-541. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 3.     | S.Muthu Sanjay    | Eco-friendly light weight bricks                                           | CTM (2000 kN capacity),                                    | Got fund for the project under TNSCST Students Project Scheme                                                                                                                        | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 4.     | K. Jeevanbabu     | Figure Diminishing Geopolymer products                                     | CTM (2000 kN capacity),                                    | Funded project Granted in the scheme of MSME Idea Hacathon                                                                                                                           | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 5.     | M.Paartha Sarathi | Experimental Study on Paver Block using Prosopis Juliflora Ash             | CTM (2000 kN capacity),UTM, Vicat and Lechatlier Apparatus | Presented at International Conference on Sustainable Technology in Civil Engineering and Applied Sciences 2023 (ICSTCA2023)& Published in the conference of Global Nest Journal      | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 6.     | V.Vineeth         | Experimental Study on mechanical properties of Textile Reinforced Concrete | CTM (2000 kN capacity), UTM                                | Presented at International Conference on Sustainable Technology in Civil Engineering and Applied Sciences 2023 (ICSTCA2023' & Published in the journal of E3 Web of conferences      | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl.No. | Team Members                                                 | Details of the Project                                                        | Facilities Utilized                                                                 | Outcome of the projects                                                                                                                                                                                                                                          | Achievements of PO/PSO                                                                |
|--------|--------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 7.     | K. Hariharasudhan                                            | Treatment of textile effluent in palm fruit cell                              | COD analyser, Muffle Furnace, Digital Turbidity meter, pH meter, Conductivity Meter | Subha, C. & Hariharasudhan, K. (2024). Textile Waste Water Treatment Using Biochar. Journal of Science Technology and Research (JSTAR) 5 (1):547-551.                                                                                                            | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 8.     | S.Ajandha Devi<br>G.Jenitha<br>M.Subalakshmi                 | Numerical study of flexural behavior of reinforced concrete beam using MATLAB | MAT Lab                                                                             | Presented at the International Conference on Smart Technologies and Applications 2022 and accepted for publication in AIP Conference Proceedings                                                                                                                 | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 9.     | S.Ajandha Devi<br>G.Jenitha<br>M.Subalakshmi                 | Experimental investigation on packing density of different fine aggregate     | Sieve analysis, CTM 2000kN capacity                                                 | Indhumathi, M. ; Leema Margret, A. ; Ajandhadevi, S. ; Jenitha, G. & Subalakshmi, M. (2024). Experimental Investigation on Packing Density of Different Fine Aggregate In Geopolymer Concrete. Journal of Science Technology and Research (JSTAR) 5 (1):552-566. | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 10.    | B.Jacksingh<br>Dharma<br>K.Thanga Guru<br>P.Karthick<br>Raja | Experimental and Numerical Studies on Gypsum Board using ABAQUS               | Numerical analysis done by Abaqus software, Flexure Testing Machine                 | Presented at the International Conference on Smart Technologies and Applications 2022                                                                                                                                                                            | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 11.    | A.Abdhur Rahman<br>K.Dinesh                                  | Experimental study on Microbial fuel cell performance                         | BOD and COD analyser, Muffle furnace, pH meter                                      | ICISDG'23: International Conference on Innovation Towards Sustainable Development Goals                                                                                                                                                                          | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1,                  |

| Sl.No. | Team Members                                         | Details of the Project                                                | Facilities Utilized | Outcome of the projects                                       | Achievements of PO/PSO                                                           |
|--------|------------------------------------------------------|-----------------------------------------------------------------------|---------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------|
|        |                                                      | on pollutant removal by utilizing sesame oil cake as Biochar catalyst |                     |                                                               | PSO2, PSO3, PSO4                                                                 |
| 12.    | G.Mohanrajan<br>S.Srivarshan<br>M.RA.Suresh<br>Kumar | Experimental Study of Performance of Cement-Based Batteries           | Voltmeter           | Got fund for the project under TNSCST Students Project Scheme | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

Students and Faculty members are insisting to do their project work and research work in project laboratory. They presented their work in seminar, symposium and published in journal papers. Some of the projects are got funding from funding agencies. Some of the Important projects are listed below:

| Sl. No. | Team Members                                                     | Details of the Project                                                        | Facilities Utilized                       | Outcome of the projects                                                                                                                     | Achievements of PO/PSO                                                           |
|---------|------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1.      | Anto Vasanthan K<br>Ayyanar G<br>Karthik Raja M<br>Muneeswaran M | Experimental Study on Concrete by Partial Replacement of Sand With Bottom Ash | "Compression Testing Machine, Slump Cone" | "Presented in National Conference on Innovations in Engineering, Science and Technology NCIEST, Ramco Institute of Technology, Rajapalayam" | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl. No. | Team Members                                            | Details of the Project                                                                            | Facilities Utilized                            | Outcome of the projects                                                                                                                                                                                                                         | Achievements of PO/PSO                                                           |
|---------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 2.      | Sheelarani T,<br>Vishalini K,<br>Yamini S               | Experimental Investigation on Properties of Pervious Concrete                                     | CTM, Los Angeles Abrasion Testing Machine      | Presented a conference paper titled "Laboratory Study of Porosity and Strength Properties of Pervious Concrete" in National Conference on Innovations in Engineering, Science and Technology NCIEST, Ramco Institute of Technology, Rajapalayam | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 3.      | Keerthana C,<br>Pavithra S,<br>Subhalakshmi L V         | Experimental Study on Induction Concrete                                                          | CTM, Furnace and Oven Systems, Thermometer     | Presented in National Science day Projects                                                                                                                                                                                                      | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 4.      | Dhanalakshmi S,<br>Jaya Priya Bharathy R,<br>Ponpriya M | Android Application For Agriculture                                                               | CAD lab Systems for developing the application | won Second Prize Rs. 50,000 from Dr.A.P.J.Abdul Kalam Innovative Ecosystems under the category of Agriculture at Sri Sairam Engineering College, Chennai on 08.04.2017.                                                                         | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 5.      | Anusuya R<br>Chandraleka D<br>Sarumathi S               | Experimental study on concrete by replacement of fine aggregate with copper slag, GGBS and M-Sand | CTM,                                           | "Published a journal International Journal of Innovative Science and Research Technology in Volume 3, Issue 3, March– 2018"                                                                                                                     | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl. No. | Team Members                                              | Details of the Project                                                                                       | Facilities Utilized                                                                 | Outcome of the projects                                                                                                                                                                                                                             | Achievements of PO/PSO                                                           |
|---------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 6.      | Arun Priyan T<br>Ganesh S<br>Gnanaraj S<br>Gopal Raj S    | Reduction of fluoride content in ground water using papaya leaf charcoal and creating an android application | pH meter, Conductivity meter, TDS meter, Muffle Furnace, BOD Analyser, COD Analyser | 1. Won First Prize in the National Science Day Project Contest held at college level on 28.02.20218 2. Got a fund of Rs 7,500 under TNSCST Students Project Scheme.                                                                                 | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 7.      | Deepa Jothi P<br>Keerthana A R<br>Latha M<br>Nandhini R   | Behaviour of plate girder bridge with and without retrofitting                                               | "Deflection measurement device"                                                     | "Presented a conference paper titled ""Behaviour of Plate Girder Steel Bridge with and without Retrofitting"" in National Conference on Sustainability in Construction – NACSIC 2018, Kalasalingam Academy of Research and Education, Krishnankoil" | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 8.      | Loga Ganesh S<br>Mahalakshmi S<br>Muthu Selvi R           | Effect of specimen size on mechanical properties of pervious concrete                                        | CTM                                                                                 | "Presented a conference paper titled ""Shape and Size effect on Mechanical Properties of Pervious Concrete" in National Conference on Sustainability in Construction – NACSIC 2018, Kalasalingam Academy of Research and Education, Krishnankoil."  | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 9.      | Chinthamani.M,<br>Daniel G,<br>Ganapathi<br>Subramaniam M | Power generation from wastewater by Microbial Fuel Cell                                                      | pH meter, COD Analyser. Muffle                                                      | Presented a paper titled Power Production from Wastewater By                                                                                                                                                                                        | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11,                              |

| Sl. No. | Team Members                                                     | Details of the Project                                              | Facilities Utilized | Outcome of the projects                                                                                                                                                                                                                                    | Achievements of PO/PSO                                                           |
|---------|------------------------------------------------------------------|---------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|         |                                                                  |                                                                     | Furnace, TDS meter  | Microbial Fuel Cell in one Day National Conference on Sustainability in Construction - NASIC 2018 organized by Kalasalingam Academy of Research and Education, Krishnankoil, 13.04.2018                                                                    | PO12, PSO1, PSO2, PSO3, PSO4                                                     |
| 10.     | Amrutha P K,<br>Anisa Shifana D R,<br>Jeyasneha J,<br>Sharatha P | Effect of silica fume on mechanical properties of Pervious Concrete | CTM                 | Presented a Conference paper titled "Effect of Silica Fume on Mechanical Properties of Pervious Concrete" National Conference on Recent Trends in Civil Engineering (NCRTE'19), Mother Teresa College of Engineering and Technology, Pudukottai, Tamilnadu | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 11.     | Dharini R,<br>Gokula Nandhini G                                  | Experimental Study on Utilization of Palm Fibre in Concrete         | CTM                 | Presented a paper titled " Experimental Study on Utilization of Palm Fibre in Concrete" in a National Level Conference on Emerging Trends in Engineering Design & Manufacturing on 23.06.2019                                                              | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |



| Sl. No. | Team Members                                        | Details of the Project                                                                                  | Facilities Utilized                                            | Outcome of the projects                                                                                                                                                                                                                                                                    | Achievements of PO/PSO                                                           |
|---------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 12.     | Karankumar M, Sujith S                              | Structural rehabilitation and strengthening of column using micro concrete and additional reinforcement | Staad Pro, E tabs                                              | "Presented a conference paper titled ""Structural Rehabilitation and strengthening of column using Micro Concrete and additional reinforcement", National conference on Emerging Trends in Engineering Design and Manufacturing, Hindusthan Institute of Technology, Coimbatore"           | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 13.     | Raghul T, Ramkumar K, Vijaya Kumar K, Ajaykumar E.R | An eco friendly solution for sewage water treatment using natural herbs and Ag nanoparticles            | pH meter, TDS meter, Conductivity Meter, Oven, Muffle Furnace, | "Presented a Conference paper titled ""An eco-friendly solution for sewage water treatment using Natural herbs and Ag nanoparticles"" International conference on advanced topics in interdisciplinary research in Chemistry and Physics, Sri. S. Ramasamy Naidu Memorial College, Sattur" | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl. No. | Team Members                                                                    | Details of the Project                                               | Facilities Utilized | Outcome of the projects                                                                                                                                                                                                                                             | Achievements of PO/PSO                                                           |
|---------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 14.     | Ram Prasad A,<br>Selvakumar T,<br>Shiva Surya<br>Vignesh M.N,<br>Suryaprakash M | Experimental study on light weight aggregate in concrete             | CTM                 | Published paper titled, " Experimental Study on Light Weight Aggregate in Concrete" , International Journal of Engineering Research & Technology (IJERT), ISSN: 2278-0181,ETEDM - 2019 Conference Proceedings, Volume 7, Issue 06, DOI : 10.17577/IJERTCONV7IS06010 | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 15.     | Aarthi T C R,<br>Madhumitha M,<br>Muthuselvi K                                  | Effect of cement content and chemical admixture on pervious concrete | CTM                 | Presented a conference paper titled "Effect of Chemical Admixture on Pervious Concrete Properties" International Web Conference on Smart Engineering Technologies-2020, Ramco Institute of Technology, Rajapalayam. (ISBN: 978-93-5407-648-0)                       | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |
| 16.     | Abinaya<br>Parasakthi P,<br>Kaleeswari M,<br>Priyatharshani M,<br>Sivapriya M   | Pervious concrete pavement block for Rain Water Harvesting           | CTM                 | Presented a conference paper titled "Effect of Wire Mesh on Properties of Pervious Concrete Pavement Blocks" International Web Conference on Smart Engineering Technologies-2020, Ramco Institute of Technology, Rajapalayam. (ISBN: 978-93-5407-648-0)             | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

| Sl. No. | Team Members                                             | Details of the Project                                                                             | Facilities Utilized                                            | Outcome of the projects                                                                                                                  | Achievements of PO/PSO                                                           |
|---------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 17.     | Thamotharan A,<br>Arunachalaeswaran K,<br>Ramesh Kumar R | Micro pores activated carbon production from waste ANANAS COMOSUS crown for water hardness removal | pH meter, TDS meter, Conductivity Meter, Oven, Muffle Furnace, | Won Second Prize in the National Science Day Competition held at college level at Ramco Institute of Technology, Rajapalayam, 28.02.2020 | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3, PSO4 |

## 6.5 Safety measures in laboratories: (10)

| Sl.No. | Name of the Laboratory                      | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | CONCRETE AND HIGHWAY ENGINEERING LABORATORY | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> </ul> <p><b><u>General Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Standard Operating Procedure is strictly followed for all the equipment.</li> <li>✓ For emergency situations, contact numbers of Ambulance, Fire service and security officers are displayed at every floor entrance and exit point.</li> <li>✓ Machine capacity and details are displayed in each machine.</li> <li>✓ Students are instructed to follow the proper Lab dress codes.</li> <li>✓ The students are instructed not to use any equipment unless they are trained and approved as a user by the lab assistant.</li> </ul> <p><b><u>Electrical Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> <li>✓ Working space around Lab equipment and machines are maintained properly.</li> <li>✓ All electrical panels, cables and devices are thoroughly checked periodically. Earth Connections are checked periodically.</li> <li>✓ Circuit breaker is provided in each and every main supply panel. Electrical Installation is periodically monitored to avoid the leakage.</li> <li>✓ There are sufficient lightning arresters installed and maintained at various locations within the Institution.</li> <li>✓ The operation circle of the lightning arresters covers the safety of all the buildings and structures in the campus.</li> <li>✓ Proper disposal of scrap metals</li> </ul> |

| Sl.No. | Name of the Laboratory                           | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                                  | <ul style="list-style-type: none"> <li>✓ Keep pathways clear by placing extra items (books, bags, etc.) on the shelves or under the work tables. If under the tables, make sure that these items can not be stepped on.</li> <li>✓ Proper disposal of scrap metals</li> <li>✓ Provide machine guard in all equipments to avoid accidents.</li> <li>✓ Instruct the students to remove ID card during machining operations.</li> <li>✓ To allot members (3 or 4 students) in a group. So overcrowding of the machines can be avoided.</li> <li>✓ To wear leather gloves and goggles for sand moulding preparation in concrete casting process.</li> </ul> <p><b><u>Fire Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Each floor is equipped with 4 Nos of Fire Extinguishers that are placed at accessible locations.</li> <li>✓ The Fire Extinguishers are replaced periodically based on the expiry date. Fire hydrant tube and valves are placed at accessible location</li> <li>✓ Work instructions and supervisions are provided.</li> <li>✓ Effective Safety arrangements are made with multiple exists and ventilation/exhausts in laboratories.</li> </ul> <p><b><u>Computer Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ Uninterrupted power supply is provided to the computers in the laboratories.</li> <li>✓ Regular backup of data and information is taken.</li> <li>✓ Protective antivirus is installed in all the computers.</li> </ul> <p><b><u>House Keeping:</u></b></p> <ul style="list-style-type: none"> <li>✓ Equipment, staff bench, computers and other work areas are regularly cleaned.</li> <li>✓ Working area and equipment are cleanly maintained and organized.</li> <li>✓ Leave your work station clean and in good order before leaving the laboratory.</li> </ul> <p><b><u>Green Campus &amp; Building Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ The campus is certified with green campus certificate and the safety of the building is certified by the authorities of the Government of Tamilnadu.</li> </ul> |
| 2      | COMPUTER AIDED<br>BUILDING DRAWING<br>LABORATORY | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> </ul> <p><b><u>General Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Standard Operating Procedure is strictly followed for all the equipment.</li> <li>✓ To sit the chair comfortably during operate the systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Sl.No. | Name of the Laboratory | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                        | <ul style="list-style-type: none"> <li>✓ To provide well air conditioning to improve performance of the systems.</li> <li>✓ To ask the students away from electric control panels.</li> <li>✓ To remove their shoes outside.</li> <li>✓ To give proper space to move from one location to another location in the laboratory area.</li> <li>✓ To provide proper workspace for creating comfortability and good ambience.</li> <li>✓ To close the glass doors and curtains to improve the efficiency of air coolers.</li> <li>✓ Provide exit way for emergency purpose.</li> <li>✓ Be aware of the various experiment controls (start button, stop button, speed control) for each lab.</li> <li>✓ To educate the students to use all accessories in proper way in this laboratory.</li> <li>✓ For emergency situations, contact numbers of Ambulance, Fire service and security officers are displayed at every floor entrance and exit point.</li> <li>✓ Students are instructed to follow the proper Lab dress codes.</li> <li>✓ Ensure ergonomic furniture to prevent musculoskeletal issues. Chairs should be adjustable and support good posture.</li> <li>✓ Workstations have adequate space to avoid overcrowding.</li> </ul> <p><b><u>Electrical Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> <li>✓ Working space around Lab equipment and machines are maintained properly.</li> <li>✓ All electrical panels, cables and devices are thoroughly checked periodically. Earth Connections are checked periodically.</li> <li>✓ Circuit breaker is provided in each and every main supply panel. Electrical Installation is periodically monitored to avoid the leakage.</li> <li>✓ There are sufficient lightning arresters installed and maintained at various locations within the Institution.</li> <li>✓ The operation circle of the lightning arresters covers the safety of all the buildings and structures in the campus.</li> <li>✓ Proper disposal of scrap metals</li> <li>✓ Keep pathways clear by placing extra items (books, bags, etc.) on the shelves or under the work tables. If under the tables, make sure that these items can not be stepped on.</li> </ul> |

| Sl.No. | Name of the Laboratory | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                        | <p><b><u>Fire Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Each floor is equipped with 4 Nos of Fire Extinguishers that are placed at accessible locations.</li> <li>✓ The Fire Extinguishers are replaced periodically based on the expiry date. Fire hydrant tube and valves are placed at accessible location</li> <li>✓ Work instructions and supervisions are provided.</li> <li>✓ Effective Safety arrangements are made with multiple exists and ventilation/exhausts in laboratories.</li> </ul> <p><b><u>Computer Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ Uninterrupted power supply is provided to the computers in the laboratories.</li> <li>✓ Regular backup of data and information is taken.</li> <li>✓ Protective antivirus is installed in all the computers.</li> </ul> <p><b><u>House Keeping:</u></b></p> <ul style="list-style-type: none"> <li>✓ Equipment, staff bench, computers and other work areas are regularly cleaned.</li> <li>✓ Working area and equipment are cleanly maintained and organized.</li> <li>✓ Leave your work station clean and in good order before leaving the laboratory.</li> </ul> <p><b><u>Green Campus &amp; Building Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ The campus is certified with green campus certificate and the safety of the building is certified by the authorities of the Government of TamilNadu.</li> </ul> |
| 3      | Surveying Laboratory   | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> </ul> <p><b><u>General Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Standard Operating Procedure is strictly followed for all the equipment.</li> <li>✓ For emergency situations, contact numbers of Ambulance, Fire service and security officers are displayed at every floor entrance and exit point.</li> <li>✓ Students are instructed to follow the proper Lab dress codes.</li> <li>✓ The students are instructed not to use any equipment unless they are trained and approved as a user by the lab assistant.</li> <li>✓ The students are instructed that if an instrument or piece of equipment fails during use, or isn't operating properly, report the issue to a technician right away. Never try to repair an equipment problem on their own.</li> </ul> <p><b><u>Electrical Safety:</u></b></p>                                                                                                                                                                                                                                                                                                                                                                    |

| Sl.No. | Name of the Laboratory | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|        |                        | <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> <li>✓ Working space around Lab equipment and machines are maintained properly.</li> <li>✓ All electrical panels, cables and devices are thoroughly checked periodically. Earth Connections are checked periodically.</li> <li>✓ Circuit breaker is provided in each and every main supply panel. Electrical Installation is periodically monitored to avoid the leakage.</li> <li>✓ There are sufficient lightning arresters installed and maintained at various locations within the Institution.</li> <li>✓ The operation circle of the lightning arresters covers the safety of all the buildings and structures in the campus.</li> <li>✓ Proper disposal of scrap metals</li> <li>✓ Keep pathways clear by placing extra items (books, bags, etc.) on the shelves or under the work tables. If under the tables, make sure that these items can not be stepped on.</li> <li>✓ Proper disposal of scrap metals</li> <li>✓ Provide machine guard in all equipments to avoid accidents.</li> <li>✓ Instruct the students to remove ID card during machining operations.</li> <li>✓ To allot members (3 or 4 students) in a group. So overcrowding of the machines can be avoided.</li> <li>✓ To wear leather gloves and goggles for sand moulding preparation in concrete casting process.</li> </ul> <p><b><u>Fire Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Each floor is equipped with 4 Nos of Fire Extinguishers that are placed at accessible locations.</li> <li>✓ The Fire Extinguishers are replaced periodically based on the expiry date. Fire hydrant tube and valves are placed at accessible location</li> <li>✓ Work instructions and supervisions are provided.</li> <li>✓ Effective Safety arrangements are made with multiple exists and ventilation/exhausts in laboratories.</li> </ul> <p><b><u>Computer Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ Uninterrupted power supply is provided to the computers in the laboratories.</li> <li>✓ Regular backup of data and information is taken.</li> <li>✓ Protective antivirus is installed in all the computers.</li> </ul> <p><b><u>House Keeping:</u></b></p> <ul style="list-style-type: none"> <li>✓ Equipment, staff bench, computers and other work areas are regularly cleaned.</li> </ul> |

| Sl.No. | Name of the Laboratory           | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| 4      | Strength of Materials Laboratory | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> <li>✓ <b><u>General Safety Measures:</u></b></li> <li>✓ Regularly check the condition of the testing machines to avoid undue accidents.</li> <li>✓ Calibrate the equipments and checking with standards</li> <li>✓ To store the destructive test specimens in separate place and disposed off in regular interval.</li> <li>✓ To check the hydraulic oil level of equipments for smoothness of its operations.</li> <li>✓ The students are instructed to follow the proper procedures for disposing lab waste.</li> <li>✓ To instruct the students to wear goggles, glosses, proper cotton clothes to handle cast metal in stir casting equipments.</li> <li>✓ To check the inert gas level in stir casting to find any leakages.</li> <li>✓ To check the furnace temperature level and switched off when the temperature level reaches its level.</li> <li>✓ Instruct the students to remove ID card during machining operations.</li> <li>✓ Proper disposal of scrap metals.</li> <li>✓ The students are instructed that if an instrument or piece of equipment fails during use, or isn't operating properly, report the issue to a technician right away. Never try to repair an equipment problem on their own.</li> </ul> <p><b><u>Electrical Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> <li>✓ Working space around Lab equipment and machines are maintained properly.</li> </ul> |



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| Sl.No. | Name of the Laboratory    | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|        |                           | <ul style="list-style-type: none"> <li>✓ To give enough ventilation to the laboratory.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5      | Soil Mechanics Laboratory | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> </ul> <p><b><u>General Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Regularly check the condition of the testing machines to avoid undue accidents.</li> <li>✓ Calibrate the equipments and checking with standards</li> <li>✓ To store the destructive test specimens in separate place and disposed off in regular interval.</li> <li>✓ To check the hydraulic oil level of equipments for smoothness of its operations.</li> <li>✓ The students are instructed to follow the proper procedures for disposing lab waste.</li> <li>✓ To instruct the students to wear goggles, glosses, proper cotton clothes to handle cast metal in stir casting equipments.</li> <li>✓ To check the inert gas level in stir casting to find any leakages.</li> <li>✓ To check the furnace temperature level and switched off when the temperature level reaches its level.</li> <li>✓ Instruct the students to remove ID card during machining operations.</li> <li>✓ Proper disposal of scrap metals</li> <li>✓ To give enough ventilation to the laboratory</li> <li>✓ The students are instructed that if an instrument or piece of equipment fails during use, or isn't operating properly, report the issue to a technician right away. Never try to repair an equipment problem on their own.</li> </ul> <p><b><u>Electrical Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> <li>✓ Working space around Lab equipment and machines are maintained properly.</li> </ul> |

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| Sl.No. | Name of the Laboratory                                                         | Safety Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 6      | Environmental Engineering Laboratory/Water and Waste water Analysis Laboratory | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> </ul> <p><b><u>General Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Avoid wearing any costly things.</li> <li>✓ Wear gloves when using any hazardous or toxic agent.</li> <li>✓ Wash your hands before you leave the lab.</li> <li>✓ You must wear a lab coat (and do it up) in labs.</li> <li>✓ Footwear must completely cover the foot and heel (no sandals, open-toed footwear, etc.).</li> <li>✓ Do not return used chemicals back to the stock container.</li> <li>✓ Do not heat, measure or mix any chemicals in front of your face.</li> <li>✓ Many chemicals (solid, liquid or vapour) are poisonous. Do not taste chemicals. If it is necessary to smell a chemical, do so by fanning the vapours towards your nose. Never inhale directly. Avoid inhaling dust or fine powders. Use fume hoods and personal protective equipment when necessary.</li> <li>✓ Maintain clean and orderly laboratories and work area. Discard immediately unwanted items. Leave coats, books and note books in the lecture room. Make sure all spilled liquids are wiped up immediately.</li> <li>✓ Be aware of the various experiment controls (start button, stop button, speed control) for each lab.</li> <li>✓ Do not leave experiments running unattended.</li> <li>✓ Be mentally alert, always read the safety instructions and pay attention to safety signs.</li> <li>✓ Ask lab instructors if you are not sure about what to do.</li> <li>✓ Eye wash stations, emergency showers, fire extinguishers, and exits are always unobstructed and accessible.</li> <li>✓</li> </ul> <p><b><u>Electrical Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> </ul> |

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| 7      | Fluid Mechanics & Machinery Laboratory/Hydraulic Engineering Laboratory | <p><b><u>First Aid:</u></b></p> <ul style="list-style-type: none"> <li>✓ First Aid kits are placed in the laboratory. It is maintained and replenished periodically Necessary First Aid training is provided to the lab instructors.</li> </ul> <p><b><u>General Safety Measures:</u></b></p> <ul style="list-style-type: none"> <li>✓ Regularly check voltage and current during running of the components.</li> <li>✓ Regularly check the condition of the testing machines to avoid undue accidents.</li> <li>✓ To check the lubricating oil level periodically to eliminate wear and tear of the parts of the machines.</li> <li>✓ The machines are cleaned after every machining operation is over.</li> <li>✓ Calibrate the equipments and checking with standards.</li> <li>✓ To store the destructive test specimens in separate place and disposed off in regular interval.</li> <li>✓ To check the hydraulic oil level of equipments for smoothness of its operations.</li> <li>✓ The students are instructed to follow the proper procedures for disposing lab waste.</li> <li>✓ To instruct the students to wear goggles, glosses, proper cotton clothes to handle cast metal in stir casting equipments.</li> <li>✓ To maintain cleanliness in equipment's to avoid spillages of oil and water.</li> <li>✓ Instruct the students to remove ID card during machining operations.</li> <li>✓ To give enough ventilation to the laboratory</li> <li>✓ The students are instructed that if an instrument or piece of equipment fails during use, or isn't operating properly, report the issue to a technician right away. Never try to repair an equipment problem on their own.</li> </ul> <p><b><u>Electrical Safety:</u></b></p> <ul style="list-style-type: none"> <li>✓ All electrical equipment and components are inspected and approved by competent authority.</li> <li>✓ Formal training and awareness programs are arranged on electrical appliances.</li> </ul> |

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We strive to be a model for environmental, health and safety excellence in teaching, research, extension, and the management of its facilities. In pursuit of this goal, appropriate policies and procedures must be developed and followed to ensure this community operates in an environment free from recognized hazards. Faculty, staff, and students are responsible for compliance with established policies and are encouraged to enculturate practices that ensure safety, protect health, and minimize the institution's impact on the environment.

### **Role of Laboratory Practice in Engineering Education**

Engineering education is incomplete without laboratory practice. The overall goal of engineering education is to prepare students to practice engineering and in particular to deal with the nature of problems faced by the society. The laboratory practice has been an important part of professional and engineering undergraduate education; the laboratory is an ideal place for active learning. Students learn in a real world environment, function as team members, discuss the planning of experiments, and share ideas about the analysis and interpretation of data. Most engineering instruction took place in the laboratory and it demands the active use of knowledge and skill.

### **Laboratory Safety**

- Adequate safety and hygienic conditions prevail in all places of workshop.
- Housekeeping and cleanliness of the Lab is maintained at regular intervals.
- Proper use and maintenance of laboratory equipment for laboratory safety.
- Laboratory apparatus are regularly inspected to ensure proper maintenance.
- All the Laboratory equipments and scientific instruments are positioned as per plan to ensure protection.
- Sufficient space is available for easy and free movement in the Lab.
- Proper illumination is available in the Lab.
- For the safety of Laboratory, electrical devices are periodically inspected that the electrical equipments are sure to be in good condition and any power cords are not frayed or have exposed wiring.
- For first aid, a medical unit exists to close workshop. This unit operates during college hours. For major injuries/accidents, a Hospital is located nearby to the college campus and in exigencies; its services can be availed. An ambulance is available round the clock at the campus for the same.
- Sufficient arrangement of dry sand is available at an accessible place.
- Students are advised and trained, all the
- safety details in the form of Do's & Don'ts
- An Emergency alarm is available at the workshop.

### **Safety Guidelines**

#### **- General Safety**



People who work in scientific laboratories are exposed to various hazards. Most workplaces have hazards that are well recognized (those of ordinary fire, for example) with well-defined actions to control the situation. Laboratories, however, involve a greater variety of possible hazards and some of these hazards need precautions not ordinarily encountered.

An introduction to safe practices for a variety of widely used laboratory procedures is listed below:

1. No running or jumping in a laboratory is permitted. Stored items or equipment shall not block access to the fire extinguisher(s), safety equipment, or other emergency items. Stairways, hallways, passageways/aisles and access to emergency equipment and/or exits must be kept dry and unobstructed; i.e., no storage, no equipment, phone or other wiring. No combustible material such as paper, wooden boxes, pallets, etc., shall be stored under stairwells or in hallways. Hallways shall be kept free of boxes and materials so that exits and normal paths of travel are not blocked.
2. Eating or drinking within laboratories is not permitted. In all laboratories specific office areas may be designated for food in coordination with the Safety Committee. They must be physically separated from any laboratory operations. In the specified office areas no consumables, reagents or any tools should be shared with work areas.
3. No food or beverage may be stored in the cold rooms/Laboratory refrigerators and freezers.

### **Electrical Safety**

The typical laboratory requires a large quantity of electrical power. This increases the likelihood of Electrically-related problems and hazards.

The following recommendations are basic to a sound electrical safety program in the laboratory.

1. All electrical equipment shall be properly grounded.
2. Sufficient room for work must be present in the area of breaker boxes. All the circuit breakers and the fuses shall be labelled to indicate whether they are in the "on" or "off" position, and what appliance or room area is served. Fuses must be properly rated.
3. Equipment, appliance and extension cords (junction boxes) must be in good condition and must be routinely dusted.
4. Extension cords shall not be used as a substitute for permanent wiring.
5. Electrical cords or other lines shall not be suspended unsupported across rooms or passageways. Do not route cords over metal objects such as emergency showers, overhead pipes or frames, metal racks, etc. Do not run cords through holes in walls, ceilings, doorways or windows. Do not place under carpet, rugs, or heavy objects. Do not place cords on pathways or other areas where repeated abuse can cause deterioration of insulation.
6. Multi-outlet plugs shall not be used unless they have a built-in circuit breaker. This causes overloading on electrical wiring, which will cause damage and possible overheating.
7. All building electrical repairs, splices, and wiring shall be performed by the Electrical Department.

### **Handling Glassware**

1. Glass breakage is a common cause of injuries in laboratories. Only glass in good condition should be used.
2. Clean all glassware before sending for repair. Glassware that has been in contact with infectious agents shall be disinfected before disposal or repair.
3. Protect hands with leather gloves when inserting glass tubing. Hold elbows close to the body to limit movement when handling tubing.
4. Use glassware of the proper size. Allow at least 20% free space. Grasp a three-neck flask by the middle neck, not a side neck.
5. Conventional laboratory glassware must never be pressurized or used with vacuum.



**First Aid Box**



**Fire Extinguisher**



**Fire Hydrant**

Fig 6.5.1 Safety Measures

(Link : [https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil\\_NBA\\_Criteria\\_VI.pdf](https://www.ritrjpm.ac.in/images/civil/2024-2025/NBA/Civil_NBA_Criteria_VI.pdf))